Location Models

Jakob Sverre Alexandersen
GRA4154 Supply Chain Optimization with Mathematical Programming
Lecture 3

January 28, 2026

Contents

1 Demand Allocation Model

- A country consists of a number of sales markets
- Markets receive goods from a number of factories
- The challenge is to determine how much to deliver from each factory to each market

A product is produced at three factories (A, B and C) and shipped to four markets (1, 2, 3, 4)

Capacities at the three factories:

A	**B**	**C**
100	120	110

Demand at the four markets:

1	**2**	**3**	**4**
55	60	35	50

Cost per unit for production and shipping:

	1	**2**	**3**	**4**
A	7	4	9	6
B	3	8	5	4
C	6	5	10	5

Find the optimal quantities to ship from each factory to each market

Decision variable:

- $X_{i,j} \geq 0$: quantity shipped from factory i to market j

Parameters:

- cap_i: capacity at factory i
- d_j: demand at market j
- $c_{i,j}$: unit cost from factory i to market j

Model:

$$\min \sum_i \sum_j c_{i,j} X_{i,j}$$

$$\text{s.t.}$$

$$\sum_j X_{i,j} \leq \text{cap}_i \quad \forall i \tag{1}$$

$$\sum_i X_{i,j} = d_j \quad \forall j \tag{2}$$

$$X_{i,j} \geq 0 \tag{3}$$

1. The quantity shipped from a factory must not exceed its capacity
2. The quantity shipped to a market must equal the market's demand
3. The factory must ship goods

2 Some location models

- Why do we focus on location models?
  1. The modeling methods are good examples of basic modeling techniques in mathematical modeling
  2. Location analysis is one of the big application areas for optimization / operations research

2.1 Applications of location models

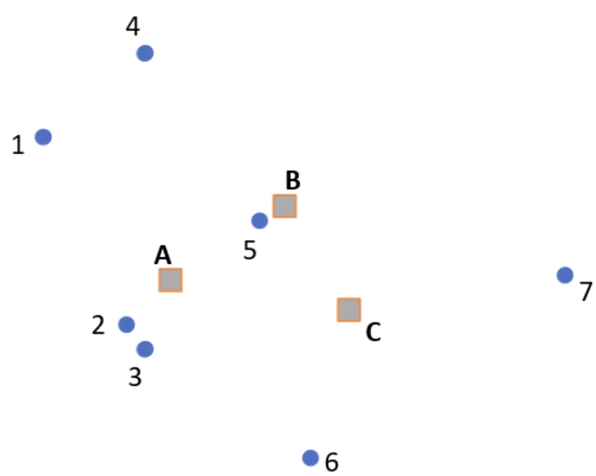
- Location of factories and/or warehouses

  "Supply network design"
- Location of public services:

  Fire stations, police stations, hospitals, welfare offices etc.

3 The P-median problem

Choose P of the candidates locations so that total weighted distance between demand nodes and the nearest facility is minimized



	A	B	C	Demand
1	35	40	59	400
2	11	35	35	700
3	15	36	33	900
4	46	38	61	800
5	18	5	23	1400
6	42	51	31	500
7	62	46	35	300

For $P = 2$, the optimal solution is to use A and B. Nodes 1, 2, 3, 6 are assigned to A. Nodes 4, 5, 7 are assigned to B. Total weighted distance: 107400 km, average distance per individual: 107400 / 5000 = 21.4 km

Decision variables:

$$X_j = \begin{cases} 1 & \text{if we locate at potential facility site } j \\ 0 & \text{otherwise} \end{cases}$$

$$Y_{ij} = \begin{cases} 1 & \text{if demands at node } i \text{ are served by a facility at node } j \\ 0 & \text{otherwise} \end{cases}$$

Parameters:

- i: index of demand node
- j: index of potential facility node
- h_i: demand at node i
- d_{ij}: distance between demand node i and potential facility site j
- P: number of facilities to be located

Model:

$$\min \sum_i \sum_j h_i d_{ij} Y_{ij}$$

$$\text{s.t.}$$

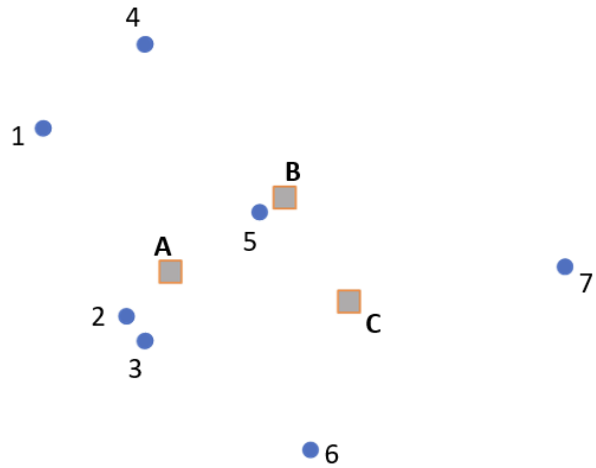
$$\sum_j Y_{ij} = 1 \quad \forall i$$

$$Y_{ij} - X_j \leq 0 \quad \forall i, j$$

$$X_j \in \{0, 1\} \quad \forall j$$

$$Y_{ij} \in \{0, 1\} \quad \forall i, j$$

4 The vertex P-center problem



	A	B	C	Demand
1	35	40	59	400
2	11	35	35	700
3	15	36	33	900
4	46	38	61	800
5	18	5	23	1400
6	42	51	31	500
7	62	46	35	300

Choose P of the candidates' locations so that **maximum** distance between any demand node and its nearest facility is **minimized**

Decision variables:

$$X_j = \begin{cases} 1 & \text{if we locate at potential facility site } j \\ 0 & \text{otherwise} \end{cases}$$

$$Y_{ij} = \begin{cases} 1 & \text{if demands at node } i \text{ are served by a facility at node } j \\ 0 & \text{otherwise} \end{cases}$$

$$D \quad \text{(maximum distance between a demand node and nearest facility)}$$

Parameters:

- i: index of demand node
- j: index of potential facility node
- h_i: demand at node i
- d_{ij}: distance between demand node i and potential facility site j
- P: number of facilities to be located

Model:

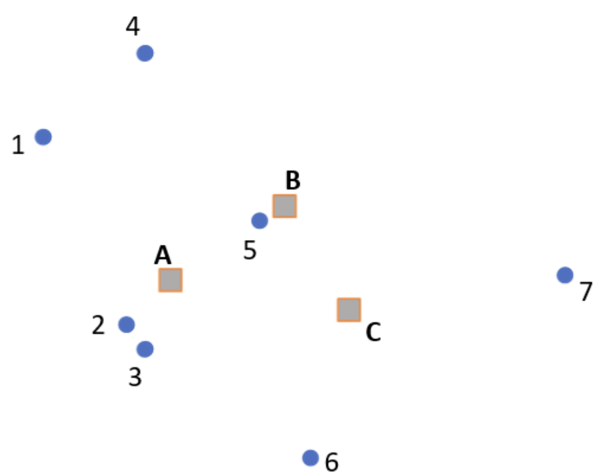
$$\min D$$
$$\text{s.t.}$$
$$\sum_j X_j = P$$
$$\sum_j Y_{ij} = 1 \quad \forall i$$
$$Y_{ij} - X_j \leq 0 \quad \forall i, j$$
$$D \geq \sum_j d_{ij} Y_{ij} \quad \forall i$$
$$X_j \in \{0, 1\} \quad \forall j$$
$$Y_{ij} \in \{0, 1\} \quad \forall i, j$$

5 The set covering problem

A demand node is said to be overed if it is within a predefined distance from the nearest facility.

For each facility, theree is **fixed cost**.

Choose the portfolio of facility locations such that all demand nodes are covered



Before the optimization, a "cover matrix" must be generated, based on a given maximum distance (here: 50 km)

Distance matrix

	A	B	C
1	35	40	59
2	11	35	35
3	15	36	33
4	46	38	61
5	18	5	23
6	42	51	31
7	62	46	35

Cover matrix (50km)

	A	B	C
1	1	1	0
2	1	1	1
3	1	1	1
4	1	1	0
5	1	1	1
6	1	0	1
7	0	1	1

Parameters:

- c_j: fixed cost of setting a facility at node j
- S: maximum acceptable service distance (or time)
- N_i: set of facility sites j within acceptable distance node i (i.e., $N_i = \{j | d_{ij} \leq S\}$)

Model:

$$\min \sum_j c_j X_j \tag{4}$$

$$\text{s.t.}$$

$$\sum_{j \in N_i} X_j \geq 1 \quad \forall i \tag{5}$$

$$X_j \in \{0, 1\} \quad \forall j \tag{6}$$

The objective function (4) minimizes the cost of facility location. In many cases, the costs c_j are assumed to be equal for all potential facility sites j, implying an objective equivalent to minimizing the number of facilities located. Constraint (5) requires that all demands i have at least one facility located within the acceptable service distance. The remaining constraint (6) require integrality for the decision variables.

6 The maximal covering problem

Specifically, the maximal covering problem seeks to maximize the amount of demand covered within the acceptable service distance S by locating a fixed number of facilities. The formulation of this problem requires the following additional set of decision variables:

$$Z_i = \begin{cases} 1 & \text{if node } i \text{ is covered} \\ 0 & \text{otherwise} \end{cases}$$

$$X_j = \begin{cases} 1 & \text{if we locate at potential facility site } j \\ 0 & \text{otherwise} \end{cases}$$

Combining these variables with the notation defined above, we derive the following formulation of the maximal covering problem:

Parameters:

- h_i: demand at node i

Model:

$$\max \sum_i h_i Z_i$$

$$\text{s.t.}$$

$$Z_i \leq \sum_{j \in N_i} X_j \quad \forall i$$

$$\sum_j X_j \leq$$

$$X_j \in \{0, 1\} \quad \forall j$$

$$Z_i \in \{0, 1\} \quad \forall i$$